Budapest University of Technology and Economics
Faculty of Natural Sciences

M1 – Transistor amplifiers

Lab report

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Abstract

We investigated two basic transistor amplifier circuits built with BC182 NPN transistors: a common-emitter (CE) amplifier and a common-collector (CC) amplifier, including the Miller effect and its suppression by a cascode connection. For the CE stage we measured the DC operating point ($U_{BE} = 0.632\,\mathrm{V}$, $U_{CE} = 6.46\,\mathrm{V}$), the saturation limit ($152\,\mathrm{mV_{pp}}$ input at 1 kHz), and the amplitude and phase response between 50 Hz and 100 kHz. The mid-band voltage gain was $A_u = 72.8$ (37.24 dB) and the phase shift 175°. With the 22 pF capacitor C_3 between base and collector (unintentionally left in the circuit during tasks c and d), the −3 dB band extended from $f_l = 66\,\mathrm{Hz}$ to $f_u = 12.58\,\mathrm{kHz}$ (bandwidth $B = 12.51\,\mathrm{kHz}$). Removing C_3 raised f_u to 82.8 kHz, and adding the second transistor (cascode) raised it to 217.8 kHz, which demonstrates the Miller effect. The CC stage had a voltage gain of 0.78, no measurable phase shift up to 100 kHz and an input resistance of 69.5 kΩ; bootstrapping the bias network increased the input resistance to 309.3 kΩ. All results are compared with simple theoretical estimates and summarised in Section 6.

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1 Introduction

Transistor amplifiers are the building blocks of analogue electronics. Depending on which electrode is common to the input and the output, three basic circuits exist: common emitter, common collector (emitter follower) and common base. Their gain, phase, input and output resistance and frequency behaviour differ, which determines where each of them is used.

The aim of this laboratory exercise was to

- set up and verify the DC operating point of a bias-divider CE amplifier,
- measure its saturation limit, frequency and phase response, cut-off frequencies and bandwidth,
- study how the collector–base capacitance degrades the high-frequency response (Miller effect) and how a cascode connection removes this limitation (task M1),
- characterise a CC amplifier (gain, phase, input resistance) and the effect of bootstrapping on its input resistance (task M2).

Section 2 summarises the theory needed, Section 3 lists the instruments, Sections 4 and 5 describe the measurements and their evaluation, and Section 6 summarises and compares all results with the expected values.

2 Theoretical background

2.1 Basic configurations

A transistor amplifier uses a small input signal to control a much larger supply power. A bipolar transistor must stay in its active region to amplify linearly: a DC bias sets the quiescent point (Q-point) so that the signal swing does not reach cut-off or saturation (class A operation). The three basic configurations are compared in Table 1; $g_m = I_C/U_{th}$ is the transconductance, $r_e = 1/g_m$ the dynamic emitter resistance ($U_{th} \approx 25.9\,\mathrm{mV}$ at room temperature) and β the current gain.

Table 1. Basic transistor amplifier configurations (small-signal, mid-band estimates).

Configuration	Voltage gain	Current gain	Phase	Typical use
Common emitter	$-g_m R_C$ (high)	$\approx \beta$	180°	audio-frequency amplification
Common collector	$\frac{R_E}{R_E + r_e} \lesssim 1$	$\approx \beta + 1$	0°	impedance matching, buffer
Common base	$+g_m R_C$ (high)	$\alpha < 1$	0°	high-frequency stages

2.2 Biasing and operating point

In the bias-divider circuit (Figure 2) the divider R_2, R_3 fixes the base voltage independently of the base current, provided the divider current is much larger than I_B:

$$U_B = U_T \frac{R_3}{R_2 + R_3}, \qquad U_E = U_B - U_{BE}, \qquad I_E = \frac{U_E}{R_4}, \qquad U_C = U_T - I_C R_C. \tag{1}$$

For a silicon transistor $U_{BE} \approx 0.5\,\mathrm{V}$ to $0.7\,\mathrm{V}$. To obtain the largest symmetric output swing, U_{CE} is set to about $U_T/2$.

2.3 Gain, cut-off frequencies and bandwidth

The voltage gain is usually given in decibels,

$$a_u = 20 \lg \left| \frac{U_{\mathrm{out}}}{U_{\mathrm{in}}} \right| \ \mathrm{dB}. \tag{2}$$

The cut-off frequencies f_l and f_u are where the gain has dropped by 3 dB, i.e. to $1/\sqrt{2} \approx 0.707$ of the mid-band value, and the bandwidth is $B = f_u - f_l$. The lower cut-off is set by the coupling and bypass capacitors, the upper one by the transistor capacitances and the Miller effect. The phase shift is obtained from the time delay t between the two signals with period T:

$$\varphi = \frac{t}{T} \cdot 360^\circ. \tag{3}$$

2.4 Miller effect and cascode connection

In an inverting stage with base-to-collector voltage gain $A_b = -g_m R_C$, the collector–base capacitance C_{CB} (and any external capacitor C_3 between these nodes) appears at the input multiplied by $(1 + |A_b|)$ [4, 2]:

$$C_M = (C_{CB} + C_3)\,(1 + g_m R_C), \qquad f_u \approx \frac{1}{2\pi R_{th}\,(C_{BE} + C_M)}, \tag{4}$$

where R_{th} is the resistance seen by the base (source, bias divider and r_π in parallel). In a cascode, a common-base transistor T_2 (base at constant DC voltage) is placed on top of the common-emitter transistor T_1. The load of T_1 is then only the small input resistance of T_2 ($\approx r_e$), so the gain of T_1 is about -1 and C_M is only about $2C_{CB}$; the voltage gain of the whole circuit is still $\approx g_m R_C$ [2, 3].

2.5 Input resistance and bootstrapping

The input resistance of the CC stage is limited by the bias network in parallel with the base input resistance $(\beta + 1)(r_e + R_E)$. It is measured by the half-output method: a series resistance is inserted in front of the input until the output voltage halves; this resistance then equals R_{in}. Bootstrapping feeds the emitter signal (which follows the base, $A_u \approx 1$) back to the lower end of a bias resistor R; the signal voltage across it shrinks to $(1 - A_u)U_{\text{in}}$, so it appears as

$$R_{\text{eff}} = \frac{R}{1 - A_u}, \tag{5}$$

which can be much larger than R [3].

3 Instruments and components

The instruments used are listed in Table 2 and shown in Figure 1. The component values of the panel are taken from the circuit diagrams of the laboratory guide [1] (Figures 2 and 5); they were not measured individually.

Table 2. List of applied instruments and components.

Instrument / part	Type	Serial no.	Pieces
Oscilloscope	Siglent SDS 1052DL+ (50 MHz, 500 MSa/s)	**[S/N]**	1
Function generator	Hameg HM 8030-4	**[S/N]**	1
Digital multimeter	Hameg HM 8011-3	**[S/N]**	1
Triple power supply	Hameg HM 8040-3 (set to +15 V)	**[S/N]**	1
Measurement panel	Transistor amplifier panel no. V, block D (M1) and block E (M2)	**[S/N]**	1
Transistors	BC182 (NPN), T_1 and T_2 on block D; T_1 on block E	–	3
Capacitor C_3	22 pF, plugged between collector and base of T_1	–	1
Jumpers (JP)	blue short-circuit caps, as set in Sections 4 and 5	–	**[no.]**
Cables	BNC and banana cables	–	**[no.]**

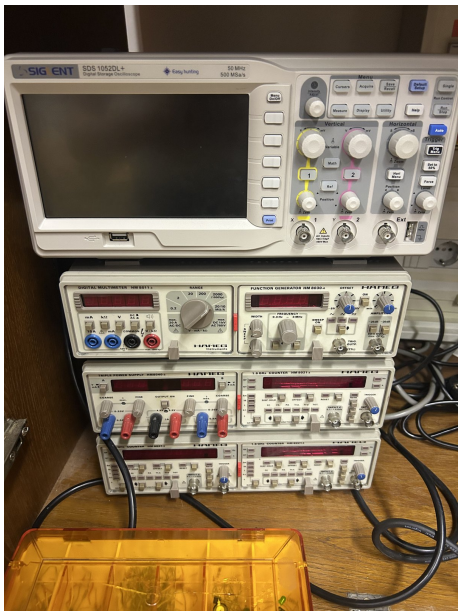


Figure 1. Measurement setup on the laboratory bench: Siglent SDS 1052DL+ oscilloscope (top), Hameg HM 8011-3 multimeter and HM 8030-4 function generator, and HM 8040-3 triple power supply. The HM 8021-3 frequency counters visible in the rack were not used.

4 M1 – Common-emitter amplifier

4.1 Circuit and set-up

The circuit under test is block D of the panel (Figure 2). Before switching on, the jumpers were set as prescribed: JP_1 open, JP_2 3–4 closed, JP_3 3–4 closed, JP_4 open. The supply was +15 V; a silicon protection diode in series with the supply lowers the voltage at the circuit to $U_T \approx 14.4\,\text{V}$ (assuming a diode drop of about 0.6 V). With JP_1 open the input signal reaches the base through $R_1 = 4.8\,\text{k}\Omega$ and C_1; R_4 is bypassed by C_2 for AC signals, and the load of T_1 is $R_9 = 7.5\,\text{k}\Omega$.

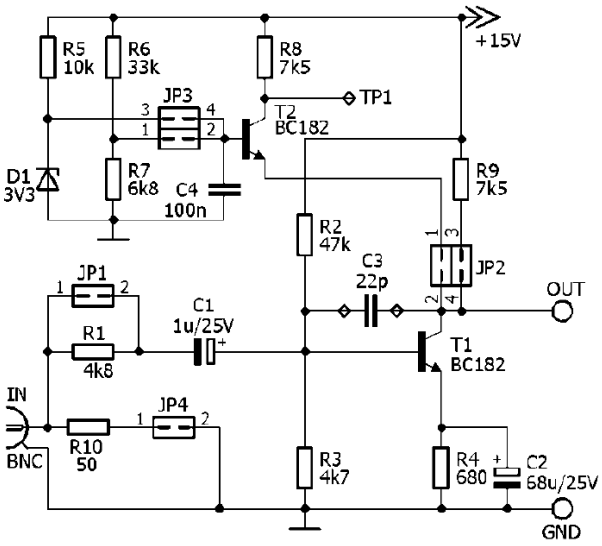


Figure 2. Common-emitter amplifier on block D of the panel (circuit diagram from the laboratory guide [1]). T_1 is the amplifying transistor, T_2 is only used in the cascode measurement (task f).

Deviation from the instructions. The guide prescribes that C_3 is not plugged in. During tasks a)–d) the capacitor was *accidentally left in the circuit*; we noticed this only afterwards. Instead of repeating the measurement series, we measured task e) with C_3 removed. Consequently, the frequency response in tasks c) and d) belongs to the amplifier *with* C_3, and the comparison without C_3 is made in task e). This does not influence the DC operating point or the mid-band gain.

4.2 a) DC operating point

With no input signal, the DC voltages of the electrodes of T_1 were measured against ground with the multimeter. The base-emitter and collector-emitter voltages follow from the differences:

$$U_{BE} = U_B - U_E = 1.288\,\mathrm{V} - 0.656\,\mathrm{V} = 0.632\,\mathrm{V},$$
$$U_{CE} = U_C - U_E = 7.12\,\mathrm{V} - 0.656\,\mathrm{V} = 6.46\,\mathrm{V}.$$

Table 3. DC operating point of T_1: measured values and expected values from Eq. (1) with $U_T = 14.4\,\mathrm{V}$.

Quantity	Measured [V]	Expected	Basis of the expectation	Deviation
U_B	1.288	1.31 V	unloaded divider R_2, R_3	−1.6 %
U_E	0.656	0.66 V	$U_B - 0.65\,\mathrm{V}$	−0.6 %
U_C	7.12	7.16 V	$U_T - I_E R_9$, $I_E = U_E/R_4$	−0.6 %
U_{BE}	0.632	0.5 V to 0.7 V	silicon junction	within range
U_{CE}	6.46	≈ 7.2 V	$U_T/2$	−10 %

Evaluation. $U_{BE} = 0.632\,\mathrm{V}$ is in the expected range for a forward-biased silicon junction, and $U_{CB} = 5.8\,\mathrm{V} > 0$ means the base-collector junction is reverse-biased, so T_1 is in the active region. $U_{CE} = 6.46\,\mathrm{V}$ is 10 % below $U_T/2$, so the operating point is close to the centre.

4.3 b) Saturation

The layout of Figure 2 of the guide was used: the function generator drives the amplifier input and channel 1 of the oscilloscope, channel 2 shows the output. At 1 kHz, the sinusoidal input amplitude was increased from $100\,\mathrm{mV_{pp}}$ until the output started to distort. The first distortion appeared at an input amplitude of

$$U_{\mathrm{in}}^{\mathrm{sat}} = 152\,\mathrm{mV_{pp}},$$

where the lower peaks of the output waveform were clipped (the transistor goes into saturation). With $A_u = 72.8$ this corresponds to an output of about $11.1\,\mathrm{V_{pp}}$.

Evaluation. The clipping appeared at the bottom of the waveform, as expected from the Q-point (less room towards saturation than towards cut-off). The Q-point allows about $12.6\,\mathrm{V_{pp}}$ output, i.e. an input of about $172\,\mathrm{mV_{pp}}$; the measured value is 12 % lower, within the 30 % limit (the onset of distortion was judged by eye).

4.4 c) Frequency response

The frequency dependence was measured with an input of $U_{\mathrm{in}} \approx 50\,\mathrm{mV_{pp}}$, much below the saturation limit and is therefore undistorted, at the frequencies of Table 4. At each frequency the output peak-to-peak voltage was read on the oscilloscope and the phase shift between input and output was determined from the time delay of the two signals (comparison method, Eq. (3)). The gain was calculated as $A_u = U_{\mathrm{out}}/U_{\mathrm{in}}$ and a_u from Eq. (2).

Table 4. Measured frequency dependence of the CE amplifier (with C_3 plugged in, $U_{\text{in}} \approx 50\,\text{mV}_{\text{pp}}$): output voltage, voltage gain A_u (ratio) and a_u (dB), and phase shift φ.

f [Hz]	U_{in} [mV]	U_{out} [mV]	A_u	a_u [dB]	φ [°]
50	50	2120	42.4	32.55	112
100	50	3120	62.4	35.90	133
200	50	3520	70.4	36.95	156
500	50	3600	72.0	37.15	169
1000	50	3640	72.8	37.24	175
2000	50	3600	72.0	37.15	183
5000	50	3400	68.0	36.65	199
10 000	50	2960	59.2	35.45	218
20 000	50	2000	40.0	32.04	235
50 000	50	960	19.2	25.67	253
100 000	50	520	10.4	20.34	254

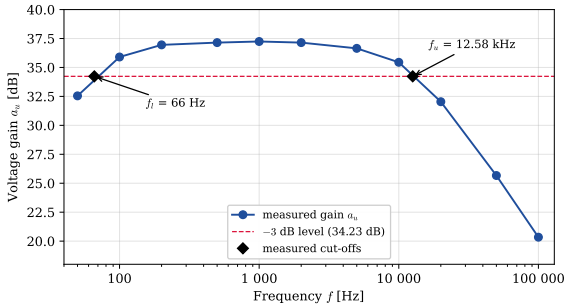


Figure 3. Amplitude characteristic (Bode plot) of the CE amplifier: voltage gain a_u versus frequency on a logarithmic axis. Dashed line is the −3 dB level below the maximum; Diamonds are the directly measured cut-off frequencies.

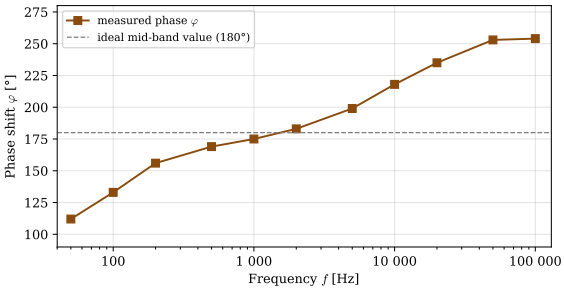


Figure 4. Phase characteristic of the CE amplifier: phase shift φ between input and output versus frequency (logarithmic axis).

Evaluation. The gain is flat around 1 kHz with a maximum of $A_u = 72.8$ ($a_u = 37.24\,\text{dB}$) and falls towards both low and high frequencies (Figure 3). The phase shift at 1 kHz is 175°, i.e. the stage inverts the signal as expected (180°); it leads at low and lags at high frequencies (Figure 4). The expected mid-band gain is about 77–112 for the datasheet range $\beta = 120 \ldots 500$ [5]; the measured value is 5 % below the lower end, within the 30 % limit.

4.5 d) Cut-off frequencies and bandwidth

The cut-off frequencies were found by changing the frequency at constant input amplitude until the output dropped to $1/\sqrt{2}$ of its mid-band value:

$$U_{cutoff} = \frac{3640\,\text{mV}}{\sqrt{2}} \approx 2574\,\text{mV} \quad (a_u = 34.23\,\text{dB}).$$

The measured values (with C_3) are

$$f_l = 66\,\text{Hz}, \qquad f_u = 12\,580\,\text{Hz}, \qquad B = f_u - f_l = 12\,514\,\text{Hz} \approx 12.51\,\text{kHz}.$$

Evaluation. The values agree with Table 4. The lower cut-off is set by the emitter bypass capacitor C_2; the expected value is about 65 Hz, which matches the measured 66 Hz. The upper cut-off is discussed in task e).

4.6 e) Effect of the capacitor C_3 (Miller effect)

Since C_3 was already plugged in during tasks c) and d), the upper cut-off without C_3 was measured after removing it. The mid-band output was unchanged (3640 mV, cut-off level 2574 mV). The results are given in Table 5.

Table 5. Upper cut-off frequency of the CE amplifier in the three configurations, with the estimate from Eq. (4) ($\beta = 120 \ldots 500$, $C_{CB} \approx 4\,\text{pF}$, $C_{BE} \approx 40\,\text{pF}$) and the gain–bandwidth product $A_u f_u$.

Configuration	f_u measured [kHz]	A_u	f_u estimated	$A_u f_u$ [MHz]
CE with $C_3 = 22\,\text{pF}$ (tasks c, d)	12.58	72.8	11 kHz to 16 kHz	0.92
CE without C_3 (task e)	82.8	72.8	70 kHz to 103 kHz	6.0
Cascode (task f)	217.8	74.4	Miller effect removed	16.2

Evaluation. Removing C_3 raises f_u from 12.58 kHz to 82.8 kHz, a factor of 6.6. This is the Miller effect: C_3 lies between base and collector of an inverting stage with a gain of about -280, so it acts at the input as a capacitance about 280 times larger and limits the bandwidth. Both measured values lie inside the estimated ranges of Table 5.

4.7 f) Cascode connection

T_2 was plugged in with JP_2 1–2 closed and 3–4 open, so that T_2 is connected in cascode on top of T_1 (its base is held at a constant DC voltage by the zener-based divider and C_4). The output was taken at the collector of T_2 (output "Ki2" / TP1 of the guide, load $R_8 = 7.5\,\text{k}\Omega$). C_3 was not plugged in. The mid-band output was 3720 mV for $50\,\text{mV}_{\text{pp}}$ input ($A_u = 74.4$, $a_u = 37.4\,\text{dB}$), so the cut-off level is $3720\,\text{mV}/\sqrt{2} \approx 2630\,\text{mV}$. The upper cut-off frequency is

$$f_u^{\text{cascode}} = 217.8\,\text{kHz}.$$

Evaluation. Compared with the single transistor without C_3, f_u increased by a factor of 2.6, and by a factor of 17.3 compared with the circuit containing C_3 (Table 5), while the mid-band gain stayed the same (74.4 versus 72.8). The reason is that T_2 is a common-base stage with its base at AC ground: it is the only load of T_1, so the gain of T_1 is only about -1 and the Miller multiplication of C_{CB} practically disappears. The remaining limit is most likely the output node (load, oscilloscope and stray capacitances).

5 M2 – Common-collector amplifier

5.1 Circuit and set-up

The circuit is block E of the panel (Figure 5). The settings before the measurement were: P_1 turned counter-clockwise to its end point, JP_1 1–2 open and 3–4 closed, JP_2 open, JP_3 open. In this configuration the base is connected directly to the divider $R_2 = R_3 = 82\,\text{k}\Omega$ and the emitter resistor is $R_5 = 1\,\text{k}\Omega$; the output is taken at the emitter.

5.2 a) Gain and phase shift at 1 kHz

A sinusoidal signal of $U_{\text{in}} = 1\,\text{V}_{\text{pp}}$ and 1 kHz was applied. The output was an undistorted sine wave in phase with the input, with $U_{\text{out}} = 780\,\text{mV}_{\text{pp}}$:

$$A_u = \frac{780\,\text{mV}}{1000\,\text{mV}} = 0.78 \quad (a_u = -2.16\,\text{dB}), \qquad \varphi = 0^\circ.$$

Evaluation. A gain below 1 and no phase shift are the expected properties of a CC stage. The expected gain is about 0.96; the measured 0.78 is 19 % lower, within the 30 % limit. The likely cause is a residual resistance of P_1 at its end point (see task c).

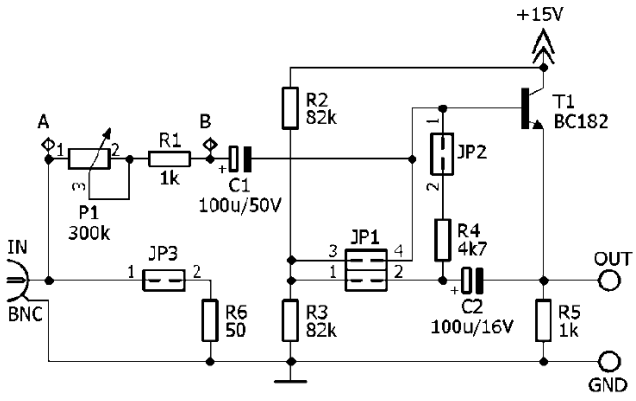


Figure 5. Common-collector amplifier (emitter follower) on block E of the panel (circuit diagram from the laboratory guide [1]). The input resistance is determined between points A and B.

5.3 b) Frequency dependence

The measurement was repeated at 100 Hz, 10 kHz, 100 kHz and 1 MHz with similar input amplitude (Table 6; the 1 kHz values are from a)).

Table 6. Common-collector amplifier: output voltage, gain and phase shift for different input frequencies ($U_{\text{in}} = 1\,\text{V}_{\text{pp}}$).

f [Hz]	100	1000	10000	100000	1000000
U_{out} [mV$_{pp}$]	780	780	780	780	780
A_u	0.78	0.78	0.78	0.78	0.78
a_u [dB]	−2.16	−2.16	−2.16	−2.16	−2.16
φ [°]	0	0	0	0	19

Evaluation. The gain is constant (0.78) from 100 Hz to 1 MHz and the phase shift is 0° up to 100 kHz; only at 1 MHz does a phase shift of 19° appear. The upper cut-off of the CC stage is therefore in the MHz range, much higher than that of the CE stage, because the voltage gain is close to 1 and the Miller multiplication of C_{CB} does not occur.

5.4 c) Input resistance

At 1 kHz the maximum output was 780 mV with P_1 at its end point. P_1 was then turned until the output halved (390 mV). After switching off the supply and removing the input signal, the resistance between points A and B was measured with the multimeter. Since this series resistance halved the output, it equals the input resistance:

$$R_{in} = 69.5\,\text{k}\Omega.$$

Evaluation. The expected input resistance is at most $R_2 \parallel R_3 = 41\,\text{k}\Omega$ (about 31 kΩ to 38 kΩ with the base resistance included), so the measured 69.5 kΩ is about 1.8 times larger, beyond the 30 % limit. We did not measure the actual resistor values or the residual resistance of P_1, so we cannot confirm the cause. The most likely explanation is that P_1 was not exactly at zero resistance at the start, so the reference output was already attenuated (this would also explain the low gain in task a)); the half-amplitude point read by eye and component tolerances also contribute.

5.5 d) Bootstrapped common-collector amplifier

P_1 was turned back counter-clockwise to its end, and the jumpers were set to JP$_1$ 1–2 closed and 3–4 open, JP$_2$ closed, JP$_3$ open (so-called high-input-resistance CC amplifier). Now the maximum

output was 864 mV ($A_u = 0.864$, $a_u = -1.27\,\mathrm{dB}$) and the half-output value 432 mV was reached with a resistance between A and B of

$$R_{in}^{\mathrm{boot}} = 309.3\,\mathrm{k\Omega}.$$

Evaluation. The input resistance increased by a factor of $309.3/69.5 = 4.45$ compared with task c), and the gain rose from 0.78 to 0.864 because the amplifier loads the signal source less. The expected value is about 108 kΩ to 351 kΩ for $\beta = 120 \ldots 500$; the measured value lies inside this range. This confirms the bootstrap principle (Eq. (5)): the emitter signal is fed back through C_2 to the bias resistor R_4, so R_4 appears much larger and the amplifier hardly loads the source.

6 Summary

All tasks of M1 and M2 were completed. The measured results and their comparison with the expected values are collected in Table 7.

Table 7. Summary of the results.

Task	Measured	Expected / comment
M1 a) U_{BE}, U_{CE}	0.632 V, 6.46 V	0.5 V to 0.7 V; $U_T/2 = 7.2$ V (−10 %)
M1 b) saturation	152 $\mathrm{mV_{pp}}$	≈ 172 $\mathrm{mV_{pp}}$ from the Q-point (−12 %)
M1 c) gain, phase at 1 kHz	$A_u = 72.8$ (37.24 dB), 175°	≈ 77 … 112 for $\beta = 120 \ldots 500$ (−5 % to −35 %); 180°
M1 d) f_l, f_u, B (with C_3)	66 Hz, 12.58 kHz, 12.51 kHz	$f_l \approx 65$ Hz; $f_u \approx 11$ kHz to 16 kHz
M1 e) f_u without C_3	82.8 kHz	≈ 70 kHz to 103 kHz; factor 6.6 above the value with C_3
M1 f) f_u, cascode	217.8 kHz	Miller effect suppressed: 2.6× (vs. e) and 17.3× (vs. d)
M2 a), b) gain, phase	$A_u = 0.78$; 0° up to 100 kHz, 19° at 1 MHz	$A_u \approx 0.96$ (−19 %); 0°
M2 c) R_{in}	69.5 kΩ	$\lesssim$ 41 kΩ (about +80 %, not fully explained)
M2 d) R_{in}, bootstrapped	309.3 kΩ	108 kΩ to 351 kΩ; 4.45× higher than c)

The CE amplifier gives a large voltage gain (72.8, 37.2 dB) with a phase inversion of about 180°, but its bandwidth is limited at high frequencies by the Miller effect: with the 22 pF capacitor C_3 the upper cut-off was only 12.58 kHz, without it 82.8 kHz, and with the cascode connection 217.8 kHz at the same mid-band gain. The CC amplifier has a voltage gain below 1, no phase shift and a flat response up to the MHz range, and its input resistance, 69.5 kΩ, can be increased by a factor of 4.45 by bootstrapping.

Most results agree with simple theoretical estimates within the uncertainty of the transistor parameters (β, C_{CB}) and of the reading of the oscilloscope. The exceptions are the input resistance in M2 c), which is about 1.8 times larger than the upper limit set by the bias divider, and the CC gain of 0.78, which is lower than expected; both are probably connected with a residual series resistance or the judgement of the half-amplitude point, which we could not verify.

Sources of error and limitations. (1) The capacitor C_3 was unintentionally left in the circuit during tasks c) and d); the data are valid for the amplifier with C_3, and task e) provides the comparison without it. (2) The value $U_C = 0.712$ V in the laboratory notes was a misplaced decimal point and was corrected to 7.12 V (see task a)). (3) The component values were taken from the nominal values of the panel and not measured, so the theoretical estimates carry the component tolerances and the unknown β of the transistors. (4) The output voltages were read with a resolution of 20 mV, the cut-off frequencies and the saturation onset were judged by eye.

References

[1] Budapest University of Technology and Economics, Electronics Laboratory (BMETE80AP00): *Measurement of basic circuits of transistor amplifiers*, laboratory guide, pp. 10/1–10/10.

[2] A. S. Sedra, K. C. Smith: *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015.

[3] P. Horowitz, W. Hill: *The Art of Electronics*, 3rd ed., Cambridge University Press, 2015.

[4] J. M. Miller: Dependence of the input impedance of a three-electrode vacuum tube upon the load in the plate circuit, *Scientific Papers of the Bureau of Standards*, vol. 15, no. 351, pp. 367–385, 1920.

[5] Central Semiconductor: *BC182 silicon NPN transistor*, datasheet, R0 (26 October 2017) (typical data for the BC182: $h_{FE} = 120 \ldots 500$ at $I_C = 2\,\mathrm{mA}$, $C_{ob} \leq 5\,\mathrm{pF}$, $f_T \geq 150\,\mathrm{MHz}$).